

Spectral Collapse and Alternating RMS Asymptotics for the Rvachev Up-Function

An exact finite-matrix and boundary-factor theorem
for analytic transfer operators

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Abstract

The Fourier-decay research in the ProveIt repository explicitly leaves open whether the exhibited eigenvalue $-1/4$ of its quadratic transfer operator is truly subleading, and whether the observed alternating RMS correction is nonzero. We resolve these questions on explicitly specified analytic Hardy spaces. More generally, for every integer dilation $b \geq 2$ and trigonometric polynomial weight w , we prove that the Fredholm determinant of the interval transfer operator is the product of a finite Fourier determinant and the explicit boundary factor $\prod_{\ell \geq 1} (1 - zw(0)b^{-\ell})$. When $w(0) = 0$, all nonzero generalized eigenspaces lie in a finite Fourier band.

For $w(x) = \sin^2(\pi x)$ and $b = 2$, the complete spectrum is $\{0, 1/2, -1/4\}$, with algebraic multiplicities one and two at the nonzero values. We derive exact recursions for the spectral projections, a quantitative remainder of order $\exp(-\frac{1}{2}(\log 2)n^2 + O(n \log n))$, and a two-term formula for the squared normalized shell RMS of the Rvachev Fourier product. An executable certificate using only integer and rational arithmetic proves that its alternating coefficient is strictly negative. In particular, every even shell index $n \geq 10$ lies below the limiting RMS and every odd index $n \geq 11$ lies above it. We also obtain complete finite-matrix spectral reductions for all even powers of the sine weight, exact fourth-moment formulas, and a high-dilation spectral-collapse corollary.

The analytic proofs are conventional mathematical proofs, not new Lean formalizations. The contribution is a resolution of an identified repository frontier and explicit extensions; priority for the general determinant specialization in the wider literature is not asserted.

Keywords. Rvachev up-function; Fabius function; transfer operator; Fredholm determinant; Thue–Morse product; boundary jets; interval arithmetic.

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1 The question, its scope, and the answer

1.1 A precise repository frontier

The relevant source is the canonical Fourier-decay synthesis in ProveIt’s Fabius-function project [1]. Its subsection *Exact transfer identities without a spectral overclaim* distinguishes explicit eigenfunctions from a complete spectral theorem. It records that $-1/4$ is known to be an eigenvalue, but that its subleading status and an alternating $(-1/2)^n$ relative RMS correction still need a spectral bound on a specified function space. This is an appropriate target: it has a precise operator, an observable with concrete asymptotic significance, and a clear criterion for completion.

The inspected source at commit `de4ac5ab1438bcbeed58481c4449b593074a1a1f` has blob identifier `aeba04f6c5f2d1eccff67645f773ba71803fa2a7`. The statement of the gap occurs at source lines 1551–1555 of the canonical TeX file. The full path and immutable link are recorded in section A. The repository was changing during the investigation; these identifiers, rather than a claim about an indefinitely current branch, fix the comparison.

There are two logically separate issues. First, an eigenfunction computation does not exclude other eigenvalues of larger modulus, or additional spectrum arising from nonperiodic analytic functions. Second, even a genuine subleading eigenvalue need not occur in the asymptotics of a particular observable. Its projection can vanish, or integration can annihilate its eigenfunction. Both issues must be settled to establish a genuine alternating RMS term.

1.2 The normalized quadratic operator

Write

$$(\mathcal{L}_2 f)(z) = \frac{1}{2} \left(\sin^2 \frac{\pi z}{2} f(z/2) + \cos^2 \frac{\pi z}{2} f((z+1)/2) \right), \quad (1.1)$$

$$\mathcal{P} = 2\mathcal{L}_2. \quad (1.2)$$

On real $x \in [0, 1]$, $\mathcal{P}$ is a positive operator satisfying $\mathcal{P}\mathbf{1} = \mathbf{1}$ and $\|\mathcal{P}f\|_\infty \leq \|f\|_\infty$. For $R > 1/2$, let H_R be the Hardy Hilbert space of the disk $D_R = \{z : |z - 1/2| < R\}$, defined precisely in section 2.

Theorem 1.1 (Quadratic spectral resolution). *On every H_R , $R > 1/2$, the operator $\mathcal{L}_2$ is trace class and*

$$\det(I - z\mathcal{L}_2) = (1 - z/2)(1 + z/4)^2. \quad (1.3)$$

Its complete spectrum is $\{0, 1/2, -1/4\}$. The nonzero eigenvalues are semisimple, with respective algebraic multiplicities one and two. Their eigenspaces are

$$\text{span}\{\mathbf{1}\}, \quad \text{span}\{u, v\}, \quad u(x) = \cos(2\pi x) + \frac{1}{3}, \quad v(x) = \sin(2\pi x). \quad (1.4)$$

Consequently, continuous linear functionals μ, a, b on H_R satisfy, for $n \geq 1$,

$$\mathcal{P}^n f = \mu(f)\mathbf{1} + (-1/2)^n (a(f)u + b(f)v) + Q^n f, \quad (1.5)$$

where Q is quasinilpotent and, for every $\varepsilon > 0$, $\|Q^n\|_{H_R \rightarrow H_R} = O_{\varepsilon, R}(\varepsilon^n)$.

Here “quasinilpotent” means spectral radius zero. It does not mean nilpotent, finite rank, or identically zero. The statement is on the stated analytic Hilbert space; it is not a claim about the spectrum on $C[0, 1]$ or $L^p[0, 1]$.

1.3 The RMS conclusion

Define the entire product and real envelope

$$\Phi(t) = \prod_{j=0}^{\infty} \frac{\sin(\pi t/2^j)}{\pi t/2^j}, \quad E(t) = \exp\left(-\frac{(\log t)^2}{2 \log 2}\right) t^{-\kappa}, \quad \kappa = \frac{\log(2\pi)}{\log 2}. \quad (1.6)$$

The value of each sinc factor at zero is one. The product is the repository's Fourier-transform normalization for the Rvachev up-function; its classical construction is reviewed in [2]. For $n \geq 1$, put

$$\mathcal{R}_n^2 = \frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} \left| \frac{\Phi(t)}{E(t)} \right|^2 dt. \quad (1.7)$$

The normalization agrees with the source's $\mathcal{M}_{2,n}$, without a shift in shell index.

Theorem 1.2 (Nonzero alternating RMS correction). *There are explicitly defined constants $M > 0$ and $B < 0$ such that*

$$\begin{aligned} \mathcal{R}_n^2 &= \frac{M}{2} + \frac{B}{6}(-1/2)^n + \epsilon_n, \\ |\epsilon_n| &\leq C_0 \exp\left(-\frac{1}{2}(\log 2)n^2 + C_1 n \log(n+1)\right). \end{aligned} \quad (1.8)$$

Here $C_0, C_1 > 0$ are constants independent of n . With $A = \sqrt{M/2}$,

$$\mathcal{R}_n = A + \frac{B}{12A}(-1/2)^n + O(4^{-n}). \quad (1.9)$$

The supplied integer-arithmetic certificate proves the strict enclosures

$$-0.002907 < B < -0.002898, \quad (1.10)$$

$$0.021221672 < M < 0.021221677, \quad (1.11)$$

$$0.10300890 < A < 0.10300894. \quad (1.12)$$

Furthermore, $\mathcal{R}_n < A$ for every even $n \geq 10$, and $\mathcal{R}_n > A$ for every odd $n \geq 11$.

The constants are $M = \mu(H)$ and $B = a(H)$ for the explicitly given analytic function H in equation (7.3). The exact coefficient definitions, the analytic remainder bounds, and the finite certificate are all included below. Thus theorem 1.2 does not infer nonvanishing from a floating-point plot.

1.4 Relation to established theory and novelty

Analytic transfer operators, periodic-orbit traces, and dynamical determinants belong to classical theory; Ruelle's work [3] is a central early reference. The trace-class determinant facts used here are standard and are stated explicitly in section 2; see [4, 5]. The relation of the sine product to the Thue–Morse measure and its Riesz products is also established mathematics [6]. Related transfer operators have been studied on continuous and L^p spaces, for example in [7]; those settings must not be silently identified with the analytic interval space used here.

The new work relative to the inspected repository consists of the complete analytic spectral proof, its boundary correction, the exact projection recursions used for certification, the quantitative remainder, and the certified nonzero correction for this specific Rvachev observable. The finite-matrix theorem below is presented with a proof rather than a global priority claim. No assertion that an internationally recognized longstanding open problem has been solved is needed or made.

2 Analytic spaces, compactness, and traces

2.1 The Hardy disk space

For $R > 1/2$, define

$$H_R = \left\{ f(z) = \sum_{k \geq 0} f_k \left(\frac{z - 1/2}{R} \right)^k : \sum_{k \geq 0} |f_k|^2 < \infty \right\}, \quad \|f\|_{H_R}^2 = \sum_{k \geq 0} |f_k|^2. \quad (2.1)$$

This definition gives a separable Hilbert space with orthonormal basis $p_k(z) = ((z - 1/2)/R)^k$. Cauchy–Schwarz yields

$$|f(z)| \leq \frac{\|f\|_{H_R}}{\sqrt{1 - |z - 1/2|^2/R^2}}. \quad (2.2)$$

In particular, restriction to $[0, 1]$ is continuous into the uniform norm. Every function holomorphic on a neighborhood of the closed disk belongs to H_R .

Let $b \geq 2$ and let w be an entire, one-periodic trigonometric polynomial. Set

$$(\mathcal{L}_{b,w}f)(z) = \frac{1}{b} \sum_{a=0}^{b-1} w\left(\frac{z+a}{b}\right) f\left(\frac{z+a}{b}\right). \quad (2.3)$$

Every inverse branch $\phi_a(z) = (z + a)/b$ satisfies

$$\left| \frac{\phi_a(z) - 1/2}{R} \right| \leq q_R := \frac{1}{b} + \frac{b-1}{2bR} < 1 \quad (z \in \overline{D}_R). \quad (2.4)$$

Lemma 2.1 (Trace-class realization). *The operator $\mathcal{L}_{b,w} : H_R \rightarrow H_R$ is trace class.*

Proof. Consider one branch $Tf = a(z)f(\phi(z))$, with a holomorphic on a neighborhood of $\overline{D}_R$ and ϕ one of the inverse branches. If $A_0 = \sup_{\overline{D}_R} |a|$, then $\|Tp_k\|_{H_R} \leq A_0 q_R^k$, since the Hardy norm is bounded by the supremum on the boundary for these holomorphic functions. The rank-one expansion

$$Tf = \sum_{k \geq 0} f_k Tp_k$$

converges in trace norm: the sum of the trace norms is at most $A_0/(1 - q_R)$. A finite sum of these branch operators is trace class. $\square$

Lemma 2.2 (Weighted affine-composition trace). *Suppose $Tf = a(z)f(\phi(z))$ is as above, ϕ is affine with slope α , and z_* is its fixed point. Then*

$$\mathrm{tr} T = \frac{a(z_*)}{1 - \alpha}. \quad (2.5)$$

Proof. Use the normalized coordinate $t = (z - 1/2)/R$, so that ϕ becomes $\psi(t) = \alpha t + \gamma$ and $|\alpha| + |\gamma| < 1$. Choose $|\gamma|/(1 - |\alpha|) < \rho < 1$. The k th diagonal coefficient of T is

$$\frac{1}{2\pi i} \int_{|t|=\rho} a(t) \psi(t)^k t^{-k-1} dt,$$

where $a(t)$ denotes the weight in the normalized coordinate. Because $|\psi(t)| < |t|$ on this contour, the geometric series is uniformly convergent there. Summing gives

$$\mathrm{tr} T = \frac{1}{2\pi i} \int_{|t|=\rho} \frac{a(t)}{t - \psi(t)} dt = \frac{a(t_*)}{1 - \alpha}.$$

The trace-class expansion in [theorem 2.1](#) justifies summing the diagonal entries. The residue formula proves the claim. $\square$

2.2 The standard spectral input

For clarity, the functional-analysis facts used subsequently are the following. A compact operator on an infinite-dimensional complex Hilbert space has only isolated eigenvalues of finite algebraic multiplicity away from zero. For a trace-class operator T , the Fredholm determinant $D_T(z) = \det(I - zT)$ is entire, its nonzero zeros are the reciprocals of nonzero eigenvalues with their algebraic multiplicities, and near zero

$$\log D_T(z) = - \sum_{n \geq 1} \frac{z^n}{n} \operatorname{tr}(T^n). \quad (2.6)$$

Riesz projections separate any finite set of isolated eigenvalues. Finally, the spectral-radius formula gives $\lim_n \|Q^n\|^{1/n} = 0$ if $\operatorname{Spec}(Q) = \{0\}$. These are standard compact-operator and trace-ideal results, not additional hypotheses about the particular sine product [4, Ch. 3]. The determinant identities are also summarized in [5, §§2–3, especially (3.1)–(3.4)].

All subsequent uses check trace class explicitly. In particular, we do not apply a trace formula to an operator on $C[0, 1]$ merely because a finite collocation matrix has a trace.

3 The exact finite-matrix and boundary-factor theorem

Write

$$w(z) = \sum_{j=-M}^M w_j e^{2\pi i j z}, \quad c = w(0) = \sum_{j=-M}^M w_j, \quad K = \left\lfloor \frac{M}{b-1} \right\rfloor. \quad (3.1)$$

The coefficients may be complex. Put $w_j = 0$ outside the displayed range, and use the notation $e_k(z) = e^{2\pi i k z}$.

Lemma 3.1 (Invariant Fourier band). *The space $V_K = \operatorname{span}\{e_k : |k| \leq K\}$ is invariant under $\mathcal{L}_{b,w}$. Its matrix, with rows and columns indexed by $-K, \dots, K$, is*

$$A_{b,w} = (w_{br-k})_{-K \leq r, k \leq K}. \quad (3.2)$$

Proof. Averaging the roots of unity gives the exact identity

$$\mathcal{L}_{b,w} e_k = \sum_{\substack{|j| \leq M \\ b|k+j}} w_j e_{(k+j)/b}. \quad (3.3)$$

If $|k| \leq K$, every resulting index has magnitude at most $\lfloor (K+M)/b \rfloor \leq K$. To check the last inequality, write $M = (b-1)K + s$ with $0 \leq s < b-1$. The coefficient of e_r is w_{br-k} , proving both assertions. $\square$

Theorem 3.2 (Exact determinant factorization). *On every H_R , $R > 1/2$, one has the entire-function identity*

$$\det(I - z\mathcal{L}_{b,w}) = \det(I - zA_{b,w}) \prod_{\ell=1}^{\infty} \left(1 - \frac{zc}{b^\ell}\right). \quad (3.4)$$

Thus the nonzero eigenvalues, counted with algebraic multiplicities, are the nonzero eigenvalues of $A_{b,w}$ together with $c/b, c/b^2, c/b^3, \dots$ when $c \neq 0$. Coincident values have their multiplicities added. Zero is also in the spectrum.

Proof. Fix $n \geq 1$ and put $B = b^n$ and

$$W_n(y) = \prod_{j=0}^{n-1} w(b^j y). \quad (3.5)$$

By iterating the inverse branches and using one-periodicity,

$$(\mathcal{L}_{b,w}^n f)(z) = \frac{1}{B} \sum_{a=0}^{B-1} W_n\left(\frac{z+a}{B}\right) f\left(\frac{z+a}{B}\right). \quad (3.6)$$

The fixed point of the a th branch is $a/(B-1)$. Applying theorem 2.2 to all branches yields

$$\mathrm{tr}(\mathcal{L}_{b,w}^n) = \frac{1}{B-1} \sum_{a=0}^{B-1} W_n\left(\frac{a}{B-1}\right). \quad (3.7)$$

Notice that both 0 and 1 occur. They are distinct interval points even though W_n is periodic.

The Fourier degree of W_n is at most $M(B-1)/(b-1)$. If $W_n = \sum_{\ell} d_{\ell} e_{\ell}$, then the coefficient of e_k in $\mathcal{L}_{b,w}^n e_k$ is $d_{(B-1)k}$. Therefore

$$\begin{aligned} \mathrm{tr}(A_{b,w}^n) &= \sum_{|k| \leq K} d_{(B-1)k} \\ &= \frac{1}{B-1} \sum_{a=0}^{B-2} W_n\left(\frac{a}{B-1}\right). \end{aligned} \quad (3.8)$$

The roots-of-unity average selects exactly those Fourier coefficients, since $|(B-1)k| \leq M(B-1)/(b-1)$ is equivalent to $|k| \leq K$ for integral k . This also covers $B-1=1$.

At the repeated endpoint, $W_n(1) = c^n$. Subtraction gives

$$\mathrm{tr}(\mathcal{L}_{b,w}^n) - \mathrm{tr}(A_{b,w}^n) = \frac{c^n}{b^n - 1} = \sum_{\ell \geq 1} \left(\frac{c}{b^{\ell}}\right)^n. \quad (3.9)$$

Use equation (2.6) near $z = 0$. Absolute convergence there permits interchanging the two sums, giving

$$\log \frac{\det(I - z\mathcal{L}_{b,w})}{\det(I - zA_{b,w})} = \sum_{\ell \geq 1} \log(1 - zc/b^{\ell}).$$

The infinite product converges locally uniformly on the entire complex plane because $\sum_{\ell} |c|b^{-\ell} < \infty$. The two entire functions agree near zero, and hence everywhere. The spectral conclusions follow from the standard determinant characterization of algebraic multiplicity. Compactness on the infinite-dimensional space supplies zero in the spectrum. $\square$

Corollary 3.3 (Boundary-zero collapse). *If $w(0) = 0$, the Fredholm determinant is a polynomial:*

$$\det(I - z\mathcal{L}_{b,w}) = \det(I - zA_{b,w}). \quad (3.10)$$

There are only finitely many nonzero spectral values. This conclusion is exact, not a limit of finite-section spectra.

Example 3.4 (Why the boundary factor cannot be deleted). For $w \equiv 1$, $K = 0$ and $A = [1]$. Nevertheless, on the analytic interval space the spectrum is

$$\{0, 1, b^{-1}, b^{-2}, \dots\},$$

not merely $\{0, 1\}$. The Bernoulli polynomials give explicit eigenfunctions: $\mathcal{L}_{b,1} B_j = b^{-j} B_j$. Indeed, their generating function $te^{tz}/(e^t - 1)$ transforms under branch averaging into $(t/b)e^{(t/b)z}/(e^{t/b} - 1)$. This verifies the eigenidentity coefficient by coefficient. The endpoint factor in equation (3.4) is exactly what accounts for these nonperiodic eigenfunctions.

4 A direct explanation of spectral collapse

The determinant argument proves the complete spectrum and multiplicities. A second argument explains why no nonperiodic generalized eigenfunction can survive when the boundary weight vanishes. This is useful both as an independent check and as a potential formalization route.

Theorem 4.1 (Generalized eigenspaces lie in the Fourier band). *Assume $w(0) = 0$ and $\lambda \neq 0$. Every $f \in H_R$ satisfying $(\mathcal{L}_{b,w} - \lambda I)^q f = 0$ for some $q \geq 1$ extends to an entire one-periodic function and belongs to V_K . Consequently its Jordan-chain structure is exactly the one in the finite matrix $A_{b,w}$.*

Proof. We use induction on the length of a generalized eigenvector chain. Let $g = (\mathcal{L}_{b,w} - \lambda I)f$; at the first step $g = 0$, and at later steps g is entire and in V_K by induction.

Entire extension. The equation $f = \lambda^{-1}(\mathcal{L}_{b,w}f - g)$ extends f from D_R to $D_{bR-(b-1)/2}$. Indeed all inverse branches map the latter disk into the former. Iteration gives radii $1/2 + b^n(R - 1/2) \rightarrow \infty$. The extensions agree on their overlaps by the original equation and the identity theorem.

Periodicity. For an entire function define $\Delta f(z) = f(z+1) - f(z)$. Reindex the inverse branches. All intermediate terms cancel, and periodicity of w gives

$$\Delta(\mathcal{L}_{b,w}f)(z) = \frac{1}{b}w(z/b)\Delta f(z/b). \quad (4.1)$$

Because g is periodic, $h = \Delta f$ satisfies

$$\lambda h(z) = \frac{1}{b}w(z/b)h(z/b).$$

If h were nonzero, let r be its order of vanishing at zero. The left side has order r , while the right side has strictly greater order because $w(0) = 0$. This is impossible. If w is identically zero the conclusion follows directly as well. Thus $h = 0$.

Finite Fourier support. An entire periodic function has a Fourier expansion $f = \sum_{k \in \mathbb{Z}} f_k e_k$, and for every fixed $\tau > 0$ there is C_τ such that $|f_k| \leq C_\tau e^{-\tau|k|}$. This follows by shifting the coefficient integral to a horizontal line. Outside $[-K, K]$, the Fourier coefficients of g vanish, so

$$\lambda f_r = \sum_{|j| \leq M} w_j f_{br-j} \quad (|r| > K). \quad (4.2)$$

Put $T_N = \sup_{|k| \geq N} |f_k|$ and $C_w = \sum_j |w_j|/|\lambda|$. For any integer $N > K$,

$$T_N \leq C_w T_{bN-M}.$$

The iterated index is $N_q = b^q(N - M/(b-1)) + M/(b-1)$ and tends to infinity geometrically. Thus $T_N \leq C_w^q C_\tau e^{-\tau N_q} \rightarrow 0$. Every Fourier coefficient outside the band vanishes. $\square$

Remark 4.2 (Boundary jets explain the geometric ladder). When $c = w(0) \neq 0$, the same difference equation shows that a nonperiodic eigenfunction with Δf vanishing to order r must have eigenvalue c/b^{r+1} . The determinant theorem proves that all these ladder values occur and gives their multiplicities, including resonances with the finite Fourier spectrum. It does not assert that resonant Jordan blocks split as a direct sum without further argument.

5 The complete quadratic spectrum and its projections

5.1 The three-dimensional calculation

For $w(x) = \sin^2(\pi x)$, the nonzero Fourier coefficients are $w_0 = 1/2$ and $w_{\pm 1} = -1/4$, and $w(0) = 0$. In the ordered basis (e_{-1}, e_0, e_1) ,

$$A_{2,w} = \begin{pmatrix} -1/4 & 0 & 0 \\ -1/4 & 1/2 & -1/4 \\ 0 & 0 & -1/4 \end{pmatrix}. \quad (5.1)$$

It follows immediately that $\det(I - zA_{2,w}) = (1 - z/2)(1 + z/4)^2$. The eigenvectors 1 , $e_1 + 1/3$, and $e_{-1} + 1/3$ are linearly independent. Their real and imaginary combinations give u and v in equation (1.4). Theorems 3.2 and 4.1 now prove all the spectral assertions in theorem 1.1.

Let Π_0 and Π_- be the Riesz projections of $\mathcal{P}$ at 1 and $-1/2$. They have the forms

$$\Pi_0 f = \mu(f)\mathbf{1}, \quad \Pi_- f = a(f)u + b(f)v.$$

They commute with $\mathcal{P}$ and annihilate one another. For $Q = \mathcal{P}(I - \Pi_0 - \Pi_-)$, compact spectral decomposition gives $\text{Spec}(Q) = \{0\}$ and equation (1.5). The spectral-radius formula then gives the asserted faster-than-geometric remainder in the Hardy norm. This argument does not assume that the operator is normal.

5.2 Positivity and the invariant probability

Proposition 5.1. *The functional μ is integration against a unique stationary probability measure on $[0, 1]$. This measure has full support. In particular, $\mu(f) > 0$ for every nonzero continuous $f \geq 0$.*

Proof. For real polynomials f , equation (1.5) converges uniformly to $\mu(f)$. Positivity and $\mathcal{P}\mathbf{1} = \mathbf{1}$ imply $|\mu(f)| \leq \|f\|_\infty$, $\mu(f) \geq 0$ when $f \geq 0$, and $\mu(1) = 1$. Polynomial density extends μ to a positive norm-one functional on $C[0, 1]$, represented by a probability measure. Uniform contraction of $\mathcal{P}$ shows, by polynomial approximation, that $\mathcal{P}^n f \rightarrow \mu(f)\mathbf{1}$ uniformly for every continuous f . Invariance $\mu(\mathcal{P}f) = \mu(f)$ follows; any stationary probability must agree with this limit and hence with μ .

For full support, let S be the nonempty closed support. If $x \in S$ and a branch probability at x is positive, the corresponding inverse-branch image belongs to S . This follows by applying stationarity to a nonnegative continuous function supported near that image. If S initially contained only endpoints, either endpoint sends positive probability to $1/2$. Thus S contains an interior point x . All inverse branches of every order then have positive probability, so S contains $(x + j)/2^n$, $0 \leq j < 2^n$. Their union is dense in $[0, 1]$. $\square$

This characterization connects the leading functional to the familiar Thue–Morse Fourier-coefficient recursion [6]. Singular continuity is not needed anywhere in the proof of the RMS theorem.

5.3 Exact coefficient recursions

The normalized operator acts on every integral Fourier mode by

$$\mathcal{P}e_{2m} = e_m, \quad \mathcal{P}e_{2m+1} = -\frac{1}{2}(e_m + e_{m+1}). \quad (5.2)$$

Define real sequences $\eta(k)$ and $\alpha(k)$ for $k \geq 0$ by

$$\begin{aligned} \eta(0) &= 1, & \eta(1) &= -\frac{1}{3}, & \alpha(0) &= 0, & \alpha(1) &= 1, \\ \eta(2m) &= \eta(m), & \eta(2m+1) &= -\frac{1}{2}(\eta(m) + \eta(m+1)), \end{aligned} \quad (5.3)$$

$$\alpha(2m) = -2\alpha(m), \quad \alpha(2m+1) = \alpha(m) + \alpha(m+1). \quad (5.4)$$

The recursive clauses are used for indices at least two, avoiding a circular definition at one. Induction proves

$$|\eta(k)| \leq 1, \quad \alpha(k) \in \mathbb{Z}, \quad |\alpha(k)| \leq k. \quad (5.5)$$

For example, $\alpha(1), \dots, \alpha(8) = 1, -2, -1, 4, -3, 2, 3, -8$.

Proposition 5.2 (Exact finite-mode reduction). *For $k \geq 1$ and $n \geq \lceil \log_2 k \rceil$,*

$$\mathcal{P}^n e_k = \eta(k) + (-1/2)^n \alpha(k)(u + iv). \quad (5.6)$$

The formula for $-k$ is the complex conjugate. Thus the expansion terminates exactly after finitely many steps for each trigonometric polynomial.

Proof. At $k = 1$, $e_1 = -1/3 + u + iv$ and $u + iv$ is an eigenfunction with value $-1/2$. For larger k , use equation (5.2) and induction. Every resulting frequency is at most $\lceil k/2 \rceil$ in absolute value. The constant and eigenfunction coefficients are precisely equations (5.3) and (5.4). The depth bound follows by iterating the halving. $\square$

If g is periodic with sufficiently rapidly decaying Fourier coefficients, these recursions calculate the projections without collocation:

$$\mu(g) = \sum_{k \in \mathbb{Z}} \hat{g}(k) \eta(|k|), \quad (5.7)$$

$$a(g) = \sum_{k \neq 0} \hat{g}(k) \alpha(|k|), \quad (5.8)$$

$$b(g) = i \sum_{k \neq 0} \operatorname{sgn}(k) \hat{g}(k) \alpha(|k|). \quad (5.9)$$

For real g , the second expression is $2 \sum_{k \geq 1} \alpha(k) \Re \hat{g}(k)$. It is essential not to assume that an arbitrary analytic function on the interval is analytically periodic. The next section provides the required endpoint regularization and also justifies these coefficient formulas for the functions used here.

6 Endpoint smoothing and an explicit supergeometric remainder

Lemma 6.1 (Gain of two matching endpoint jets). *For every function holomorphic near $[0, 1]$,*

$$\Delta(\mathcal{P}f)(z) = \sin^2(\pi z/2) \Delta f(z/2) \quad (6.1)$$

near zero. Consequently $g_r = \mathcal{P}^r f$ has matching derivatives at zero and one through order $2r - 1$.

Proof. The identity is equation (4.1) multiplied by two with $b = 2$. The factor $\sin^2(\pi z/2)$ has a zero of order two. Each iteration therefore increases the order of the endpoint difference by at least two. $\square$

Let $\Omega_\delta = \{z : \text{dist}(z, [0, 1]) \leq \delta\}$, and suppose f is holomorphic on a neighborhood of this closed set. Put $M_f = \sup_{\Omega_\delta} |f|$ and

$$C_\delta = \frac{\sinh(2\pi\delta)}{2\pi\delta}. \quad (6.2)$$

Every inverse branch maps Ω_δ into $\Omega_{\delta/2}$. Moreover,

$$\left| \sin^2(\pi z/2) \right| + \left| \cos^2(\pi z/2) \right| = \cosh(\pi \Im z).$$

Using the product identity $\prod_{j \geq 0} \cosh(t/2^j) = \sinh(2t)/(2t)$ gives, uniformly in r ,

$$\sup_{\Omega_\delta} |\mathcal{P}^r f| \leq C_\delta M_f. \quad (6.3)$$

The product identity follows by repeatedly applying $\sinh(2t) = 2 \sinh(t) \cosh(t)$ and taking the small-argument limit.

Cauchy's formula on circles of radius δ , followed by $2r$ integrations by parts, now yields

$$|\widehat{g}_r(k)| \leq \frac{C_\delta M_f (2r)!}{(2\pi\delta)^{2r} |k|^{2r}} \quad (k \neq 0). \quad (6.4)$$

All boundary terms vanish by [theorem 6.1](#). The periodic extension need only have these finitely many matching jets; no holomorphic periodic extension is asserted.

Theorem 6.2 (Quantitative remainder). *Suppose $f \in H_R$ and is holomorphic on a neighborhood of Ω_δ . For integers $n \geq r \geq 2$, put $s = n - r$. Then*

$$\begin{aligned} & \|\mathcal{P}^n f - \mu(f)\mathbf{1} - (-1/2)^n (a(f)u + b(f)v)\|_\infty \\ & \leq \frac{C_\delta M_f (2r)!}{(2\pi\delta)^{2r}} 2^{-s(2r-1)} \left(\frac{4}{2r-1} + \frac{8}{3(2r-2)} \right). \end{aligned} \quad (6.5)$$

In particular, choosing $r = \lfloor n/2 \rfloor$ for $n \geq 4$ gives

$$\|\mathcal{P}^n f - \mu(f)\mathbf{1} - (-1/2)^n (a(f)u + b(f)v)\|_\infty \leq C_{f,\delta} \exp\left(-\frac{\log 2}{2} n^2 + O_\delta(n \log n)\right). \quad (6.6)$$

Proof. For $g = g_r$, [equation \(6.4\)](#) implies absolute convergence of its Fourier series and of all three coefficient sums [equations \(5.7\) to \(5.9\)](#). Apply $\mathcal{P}^s$ termwise in the uniform norm. For $|k| \leq 2^s$ the finite-mode formula is exact. For the remaining modes, $\|\mathcal{P}^s e_k\|_\infty \leq 1$, $|\eta(k)| \leq 1$, and $\|u \pm iv\|_\infty \leq 4/3$. Their individual residuals have norm at most $2 + (4/3)2^{-s}|k|$.

Put $N = 2^s$ and $D_r = C_\delta M_f (2r)! / (2\pi\delta)^{2r}$. The total residual is bounded by

$$2D_r \sum_{k > N} k^{-2r} \left(2 + \frac{4k}{3N} \right) \leq D_r N^{1-2r} \left(\frac{4}{2r-1} + \frac{8}{3(2r-2)} \right),$$

using the integral bounds for the two decreasing power sums. For fixed $r \geq 2$, this residual and its product with 2^s tend to zero. Comparison with the unique leading and subleading terms in the Hardy-space spectral decomposition identifies the Fourier sums with $\mu(g), a(g), b(g)$. The commuting projection identities give $\mu(g) = \mu(f)$ and $(a(g), b(g)) = (-1/2)^r (a(f), b(f))$. This proves [equation \(6.5\)](#) without an unjustified interchange in the Hardy norm.

Finally, $\log((2r)!) = O(r \log r)$ and $(n-r)(2r-1) = n^2/2 + O(n)$ at $r = \lfloor n/2 \rfloor$. Substitution gives [equation \(6.6\)](#). $\square$

Remark 6.3 (The Hardy hypothesis can be removed from the scalar estimate). For a function merely holomorphic near $[0, 1]$, define the three coefficients from the absolutely convergent Fourier sums of $\mathcal{P}^2 f$, undoing the factor $(-1/2)^2$ in the last two. The same proof gives convergence and the coefficient limits. Shifting the iteration shows that the definitions are independent of the smoothing depth. Hence equation (6.5) holds for all such observables, even if they do not extend to a disk D_R . This extension concerns the uniform asymptotic estimate, not an unstated spectral theorem on a different Banach space.

7 From the Fourier product to the shell observable

7.1 Convergence and a useful complex bound

On compact subsets of $\mathbb{C}$, the factors of Φ in equation (1.6) are $1 + O(4^{-j})$. The product therefore converges locally uniformly and defines an entire function. The identity

$$\frac{\sin u}{u} = \frac{1}{2} \int_{-1}^1 e^{itu} dt$$

implies $|\sin u/u| \leq e^{|\Im u|}$. Multiplying the bounds gives

$$|\Phi(z)| \leq \exp(2\pi|\Im z|). \quad (7.1)$$

For $0 < z < 1$, every real factor is positive and the tail converges to a positive number. Thus $\Phi(z) > 0$ there, while $\Phi(1) = 0$.

7.2 Exact cancellation of the shell scale

Put $a_0 = \log 2$. Splitting the infinite product after n factors gives

$$\Phi(2^{n-1}y) = \Phi(y/2)(\pi y)^{-n} 2^{-n(n-1)/2} \prod_{j=0}^{n-1} \sin(\pi 2^j y), \quad 1 \leq y \leq 2. \quad (7.2)$$

Define

$$H(z) = \frac{\Phi((1+z)/2)^2}{\pi^2} \exp\left(\frac{\text{Log}(1+z)^2}{a_0} + (2\kappa - 2)\text{Log}(1+z)\right). \quad (7.3)$$

We use the principal logarithm. This function is holomorphic on a neighborhood of $\overline{D}_1$, since $\Re(1+z) > 0$ there. In particular $H \in H_1$. It is real and strictly positive on $[0, 1]$, with $H(1) = 0$.

Using $2^\kappa = 2\pi$, a direct simplification of equation (7.2) gives, for $x = y - 1$,

$$\left| \frac{\Phi(2^{n-1}y)}{E(2^{n-1}y)} \right|^2 = \frac{1}{2} H(x) \prod_{j=0}^{n-1} 2 \sin^2(\pi 2^j x). \quad (7.4)$$

For verification of the normalization, the powers independent of y simplify to

$$\pi^{-2n} 2^{-n(n-1)+(n-1)^2+2\kappa(n-1)} = 2^{n-1} \pi^{-2};$$

the remaining power of y is $\exp((\log y)^2/a_0 + (2\kappa - 2)\log y)$.

Proposition 7.1 (Exact transfer representation of the RMS). *For every $n \geq 1$,*

$$\mathcal{R}_n^2 = \frac{1}{2} \int_0^1 \mathcal{P}^n H(x) dx. \quad (7.5)$$

Proof. Change variables $t = 2^{n-1}y$ in equation (1.7) and use equation (7.4). The elementary branch-change identity for $\mathcal{P}$ is

$$\int_0^1 \mathcal{P}^n f(x) dx = \int_0^1 f(x) \prod_{j=0}^{n-1} 2 \sin^2(\pi 2^j x) dx.$$

It follows either by the substitutions in each inverse branch, or directly from equation (3.6) with the normalized weight. Applying it to H proves the result. $\square$

This identity is exact for the infinite sinc product. No finite-product approximation enters the analytic reduction.

8 The RMS expansion, all orders, and explicit error bounds

Put

$$M = \mu(H), \quad B = a(H), \quad A = \sqrt{M/2}. \quad (8.1)$$

By theorem 5.1, $M > 0$. Since $\int_0^1 u = 1/3$ and $\int_0^1 v = 0$, integration of equation (1.5) immediately yields

$$\mathcal{R}_n^2 = \frac{M}{2} + \frac{B}{6}(-1/2)^n + \frac{1}{2} \int_0^1 Q^n H(x) dx. \quad (8.2)$$

The remainder is faster than every geometric sequence, and theorem 6.2 gives the more explicit square-exponential bound in equation (1.8).

Corollary 8.1 (All-orders scalar expansion). *For sufficiently large n and every $\varepsilon > 0$,*

$$\mathcal{R}_n = A \left(1 + \frac{B}{3M}(-1/2)^n \right)^{1/2} + O_\varepsilon(\varepsilon^n). \quad (8.3)$$

For every fixed $J \geq 0$, this gives

$$\mathcal{R}_n = A \sum_{j=0}^J \binom{J}{j} \left(\frac{B}{3M} \right)^j (-1/2)^{jn} + O_J(2^{-(J+1)n}). \quad (8.4)$$

In particular, if $B \neq 0$,

$$(-2)^n(\mathcal{R}_n^2 - A^2) \longrightarrow B/6, \quad (-2)^n(\mathcal{R}_n - A) \longrightarrow B/(12A). \quad (8.5)$$

Proof. The expression inside the square root tends to one. On a fixed positive neighborhood of A^2 , the square-root function is Lipschitz. Thus a supergeometric error in the squared quantity remains supergeometric after taking its square root. The binomial expansion gives the finite truncations, and the first-order term gives the two limits. $\square$

8.1 A fully explicit bound for the actual observable

The certificate in the next section uses $\delta = 1/4$. We establish here the analytic constant needed both for certification and for a finite-shell sign result.

Lemma 8.2 (Uniform analytic bound). *On $\Omega_{1/4}$, $|H| < 1000$. Moreover, for every $r \geq 0$, $|\mathcal{P}^r H| < 2000$ on this set. For $g = \mathcal{P}^4 H$,*

$$|\hat{g}(k)| \leq \frac{C}{|k|^8} \quad (k \neq 0), \quad C = 3,200,000. \quad (8.6)$$

Proof. If $z \in \Omega_{1/4}$, then $\Re(1+z) \geq 3/4$, $|1+z| \leq 9/4$, and $|\Im z| \leq 1/4$. The principal logarithm $L = \text{Log}(1+z)$ satisfies $|L| < 6/5$: $|\log|1+z|| < 1$ and $|\arg(1+z)| < 1/3$ already suffice. Since $\log 2 > 2/3$, $1 < \kappa < 3$, and equation (7.1) gives $|\Phi((1+z)/2)|^2 \leq e^{2\pi|\Im z|}$, the modulus of the exponential factors in H is less than

$$\exp\left(2 + \frac{(6/5)^2}{2/3} + 4 \cdot \frac{6}{5}\right) = e^{224/25} < e^9.$$

Thus $|H| < e^9/\pi^2 < 1000$. The last elementary bound follows, for example, from $\pi > 3$ and $e < 11/4$.

Next $C_{1/4} = \sinh(\pi/2)/(\pi/2) < 2$, so equation (6.3) gives $|\mathcal{P}^r H| < 2000$. Cauchy's estimate yields $\|g^{(8)}\|_\infty \leq 8! 2000 4^8$. After eight integrations by parts,

$$|\hat{g}(k)| \leq 8! 2000 \left(\frac{2}{\pi}\right)^8 |k|^{-8} < 3,200,000 |k|^{-8},$$

using $\pi > 3$. The matching endpoint derivatives required for those integrations follow from theorem 6.1. $\square$

Taking $r = 4$ in the proof of theorem 6.2, and using this C , we obtain for every $n \geq 4$ the particularly simple bound

$$\left| \mathcal{R}_n^2 - \frac{M}{2} - \frac{B}{6}(-1/2)^n \right| \leq \frac{32C}{63} 2^{-7(n-4)}. \quad (8.7)$$

Indeed, half of the bracket $4/7 + 8/(3 \cdot 6)$ is $32/63$. Although choosing r with n gives a stronger asymptotic bound, equation (8.7) is already sufficient to certify the sign from shell ten onward.

9 A reproducible certificate for the nonzero coefficient

9.1 Reduction to finitely many Fourier coefficients

Let $g = \mathcal{P}^4 H$. Because the subleading eigenvalue of $\mathcal{P}$ is $-1/2$,

$$B = 16 a(g), \quad M = \mu(g). \quad (9.1)$$

For $K \geq 1$, truncating equations (5.7) and (5.8) and using equations (5.5) and (8.6) gives

$$\left| B - 32 \sum_{k=1}^K \alpha(k) \Re \hat{g}(k) \right| \leq \frac{16C}{3K^6}, \quad (9.2)$$

$$\left| M - \hat{g}(0) - 2 \sum_{k=1}^K \eta(k) \Re \hat{g}(k) \right| \leq \frac{2C}{7K^7}. \quad (9.3)$$

For example, the first tail is bounded by $32C \sum_{k>K} k^{-7} \leq 16C/(3K^6)$.

Let $N > 2K$, and approximate the coefficients by the exact discrete averages

$$\tilde{g}_N(k) = \frac{1}{N} \sum_{j=0}^{N-1} g(j/N) e^{-2\pi i k j / N}. \quad (9.4)$$

Absolute Fourier convergence proves the aliasing identity $\tilde{g}_N(k) = \sum_{\ell \in \mathbb{Z}} \hat{g}(k + \ell N)$. For $|k| \leq K$, therefore,

$$|\tilde{g}_N(k) - \hat{g}(k)| \leq \frac{4C}{(N - K)^8}. \quad (9.5)$$

To see this, use $|k + \ell N| \geq |\ell|(N - K)$ and $\sum_{\ell \geq 1} \ell^{-8} < 2$.

Consequently the two finite sums

$$\begin{aligned} B_{K,N} &= 32 \sum_{k=1}^K \alpha(k) \Re \tilde{g}_N(k), \\ M_{K,N} &= \tilde{g}_N(0) + 2 \sum_{k=1}^K \eta(k) \Re \tilde{g}_N(k) \end{aligned} \quad (9.6)$$

satisfy the explicit error bounds

$$|B - B_{K,N}| \leq E_B := \frac{16C}{3K^6} + \frac{64CK(K+1)}{(N-K)^8}, \quad (9.7)$$

$$|M - M_{K,N}| \leq E_M := \frac{2C}{7K^7} + \frac{4C(2K+1)}{(N-K)^8}. \quad (9.8)$$

The constants used by the program are

$$r = 4, \quad K = 128, \quad N = 512, \quad J = 28, \quad C = 3,200,000. \quad (9.9)$$

Here J is the number of sinc factors used for a sample evaluation. The rational error bounds satisfy $E_B < 3.888 \cdot 10^{-6}$ and $E_M < 1.632 \cdot 10^{-9}$.

9.2 Enclosing the infinite-product samples

For $z \in [1/2, 1]$ and $J \geq 2$, write $\Phi_J(z) = \prod_{j=0}^{J-1} \text{sinc}(\pi z/2^j)$. The elementary inequalities $1 - u^2/6 \leq \sin u/u \leq 1$ for the tail arguments, together with $\prod(1 - a_j) \geq 1 - \sum a_j$ for $0 \leq a_j \leq 1$, give

$$1 - \frac{20}{9}4^{-J} \leq \frac{\Phi(z)}{\Phi_J(z)} \leq 1. \quad (9.10)$$

At an endpoint where the finite product vanishes, the equivalent product inequality is used rather than division. The estimate follows from

$$\sum_{j=J}^{\infty} \frac{(\pi z/2^j)^2}{6} = \frac{2\pi^2 z^2}{9}4^{-J} \leq \frac{20}{9}4^{-J}.$$

Every evaluation of H uses this tail interval; the program never treats a finite sinc product as equal to the infinite one.

The inverse-branch formula needed for the samples is

$$g(x) = \sum_{\ell=0}^{15} \left(\prod_{p=0}^3 \sin^2(\pi 2^p y_\ell) \right) H(y_\ell), \quad y_\ell = \frac{x + \ell}{16}. \quad (9.11)$$

For $x = j/512$, all sample points are dyadic rationals with denominator 8192. There are exactly 8192 evaluations of H . The real parts in equation (9.6) are calculated by finite cosine sums, not by an uncertified floating-point FFT.

9.3 Integer-only elementary-function enclosures

The primary program `certify_integer.py` uses fixed-point intervals $[l2^{-100}, h2^{-100}]$ with integer endpoints. Every addition, multiplication, and division rounds its rational endpoint result outward by integer floor or ceiling. Square roots use the integer square-root algorithm. Thus the certificate requires no floating-point library.

The elementary constants and functions are enclosed as follows. Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

is evaluated with 40 alternating-series terms and the first omitted term as a remainder bound. For $|t| \leq 4$, sinc is evaluated with 20 terms of

$$\text{sinc } t = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+1)!};$$

after the first term the absolute terms decrease, so the first omitted term bounds the remainder. For $y \geq 1$, with $q = (y-1)/(y+1)$,

$$\log y = 2 \sum_{k=0}^{L-1} \frac{q^{2k+1}}{2k+1} + R_L, \quad 0 \leq R_L \leq \frac{2q^{2L+1}}{(2L+1)(1-q^2)}.$$

The sample logarithms use $L = 36$, and $\log(2\pi)$ uses $L = 128$. Finally, for $0 \leq x \leq 4$ the exponential is summed through degree 60; the positive tail is bounded by its next term divided by $1 - x/62$. All argument-range checks are assertions in the program.

These choices are deliberately generous. The limiting accuracy is set by the proven Fourier-tail estimate, not by the elementary-function precision. An independent program, `certify_constants.py`, repeats the same finite reduction with `mpmath.iv` interval arithmetic. It is a cross-check, not a dependency of the integer-only certificate.

9.4 Certificate output and the sign theorem

The integer-only run produced the following outward decimal enclosures. The JSON file also records the exact rational endpoints; the displayed decimals are obtained by integer floor and ceiling.

Quantity	Lower endpoint	Upper endpoint
B	-0.002906388035461419823	-0.002898612708277941829
M	0.021221672714945662196	0.021221675977057311570
A	0.103008913970941520154	0.103008921888002768209

These enclosures imply equations (1.10) to (1.12). The wider rational bounds, rather than any noncertified extra digits, are used in the theorem statements.

It remains to verify the promised finite-shell sign. Since $B < -0.002898$, the magnitude of its alternating term is greater than $(0.002898/6)2^{-n}$. At $n = 10$, the entirely rational inequality

$$\frac{32C}{63} 2^{-42} < \frac{0.002898}{6} 2^{-10} \quad (9.12)$$

holds. For each increment of n , the ratio of the error bound in equation (8.7) to this lower bound for the alternating term decreases by 2^{-6} . The error cannot overturn the sign of the alternating term for any $n \geq 10$. The squared RMS is therefore below A^2 at even indices and above it at odd indices. Taking square roots preserves the comparison. This completes the proof of theorem 1.2, including nonvanishing, not just an upper bound.

Remark 9.1 (What is and is not mechanically checked). The primary certificate reduces the numerical assertions to explicit integer computations and elementary series inequalities. Its execution was checked, and the independent interval implementation agreed. It is not a Lean or Rocq proof object: the Python implementation and interpreter have not been verified in a proof assistant. The analytic theorem and all tail bounds are proved in the manuscript. The supplied code and exact output make the finite numerical step independently reproducible and auditable.

10 Even powers, exact moments, and larger dilations

10.1 Every even sine power has a finite analytic spectrum

For $m \geq 1$, let $\mathcal{L}_{2m} = \mathcal{L}_{2,w_m}$, where $w_m(x) = \sin^{2m}(\pi x)$. Its Fourier coefficients are

$$(w_m)_j = \frac{(-1)^j}{4^m} \binom{2m}{m+j}, \quad |j| \leq m. \quad (10.1)$$

The expression $(-1)^j$ has its usual integral-parity meaning also for negative j .

Corollary 10.1 (Complete even-power reduction). *For every $m \geq 1$, on every H_R the nonzero spectrum of $\mathcal{L}_{2m}$ is exactly the nonzero spectrum of the $(2m+1)$ -dimensional rational matrix*

$$(A_m)_{r,k} = \frac{(-1)^{2r-k}}{4^m} \binom{2m}{m+2r-k}, \quad -m \leq r, k \leq m, \quad (10.2)$$

where the binomial coefficient is zero outside its natural range. The determinant identity and every nonzero algebraic multiplicity are exact.

Proof. Here $M = m$, $K = m$, and $w_m(0) = 0$. Apply [theorems 3.2](#) and [4.1](#) and substitute [equation \(10.1\)](#). $\square$

For the integer matrix $B_m = 4^m A_m$, exact algebra gives

$$\begin{aligned} \det(YI - B_1) &= (Y-2)(Y+1)^2, \\ \det(YI - B_2) &= (Y-1)^2(Y+4)(Y^2-2Y-16), \\ \det(YI - B_3) &= (Y+1)^2(Y^2+8Y-64)(Y^3-10Y^2-96Y+512). \end{aligned} \quad (10.3)$$

The accompanying algebra script checks these and the corresponding polynomials for $m = 4, 5$. The formulas in [equation \(10.3\)](#) are finite consequences of the displayed matrices; they are not decimal eigenvalue fits.

Proposition 10.2 (A universal outer-row factor). *For every $m \geq 1$, $\det(YI - B_m)$ is divisible by $(Y - (-1)^m)^2$.*

Proof. The row with index $r = m$ has only one nonzero entry, at column $k = m$, equal to $(-1)^m$. Likewise the row $r = -m$ has only its diagonal entry. Expanding the characteristic determinant along these two outer rows proves the factor. This is a statement about algebraic multiplicity. It does not assert that the corresponding coordinate vectors are eigenvectors, or that all possible resonances are semisimple. $\square$

10.2 The quartic spectrum and a cancellation lesson

For $m = 2$, the complete nonzero spectrum is

$$\frac{1 + \sqrt{17}}{16}, \quad -\frac{1}{4}, \quad \frac{1 - \sqrt{17}}{16}, \quad \frac{1}{16}, \quad (10.4)$$

with multiplicity two at $1/16$ and one at the other values. The leading spectral radius is $(1 + \sqrt{17})/16$, and the next modulus is $1/4$. Nevertheless that subleading operator eigenvalue is absent from the following natural scalar moment.

Proposition 10.3 (Exact fourth moments). *Let*

$$I_n = \int_0^1 \prod_{j=0}^{n-1} \sin^4(\pi 2^j x) dx, \quad I_0 = 1.$$

Then

$$\sum_{n \geq 0} I_n z^n = \frac{1 + z/4}{1 - z/8 - z^2/16}, \quad (10.5)$$

and, with $s = \sqrt{17}$ and $r_{\pm} = (1 \pm s)/16$,

$$I_n = \frac{s + 5}{2s} r_+^n + \frac{s - 5}{2s} r_-^n. \quad (10.6)$$

Proof. The branch-change identity gives $I_n = \int_0^1 \mathcal{L}_4^n \mathbf{1}$. Starting from 1, the iterates stay in $\text{span}\{1, \cos(2\pi x)\}$. In that basis the matrix is

$$T = \begin{pmatrix} 3/8 & -1/4 \\ 1/8 & -1/4 \end{pmatrix},$$

because $\mathcal{L}_4 \mathbf{1} = 3/8 + (1/8) \cos(2\pi x)$ and $\mathcal{L}_4 \cos(2\pi x) = -1/4 - (1/4) \cos(2\pi x)$. Integration selects the first coordinate. Thus the generating function is the $(1, 1)$ entry of $(I - zT)^{-1}$, which is [equation \(10.5\)](#). Equivalently, $I_{n+2} = I_{n+1}/8 + I_n/16$, with $I_0 = 1$ and $I_1 = 3/8$. Solving that recurrence proves [equation \(10.6\)](#). $\square$

Here the $-1/4$ eigenfunction of the full quartic operator is proportional to $\sin(2\pi x)$, which integration annihilates. The scalar correction in [equation \(10.6\)](#) instead has relative ratio $(1 - \sqrt{17})/(1 + \sqrt{17})$. This example explains why the separate nonvanishing certificate for the quadratic Rvachev observable was necessary. The exact fourth-moment formula is included as a derived illustration, not as a claim of previously unknown classical moment theory.

10.3 High-dilation collapse

Corollary 10.4. *Let $m \geq 1$ and $b \geq m + 2$. On every H_R , the operator $\mathcal{L}_{b, w_m}$ has spectrum*

$$\left\{ 0, \frac{1}{4^m} \binom{2m}{m} \right\}. \quad (10.7)$$

The nonzero eigenvalue is simple. After division by this value, the powers converge to a rank-one projection faster than every geometric sequence in the Hardy operator norm.

Proof. The inequality $b - 1 > m$ gives $K = 0$, and the finite matrix is the scalar constant Fourier coefficient $4^{-m} \binom{2m}{m}$. The boundary factor vanishes because $w_m(0) = 0$. Apply the determinant theorem and compact spectral decomposition. The remaining spectral radius is zero. $\square$

This is a genuine simplification from the larger dilation. At $b = m + 1$, the finite band can already have dimension three; the strict threshold in the statement is not to be replaced by a weak inequality without analysis.

11 Interpretation, limitations, and a formalization route

11.1 What has been resolved

The operator in the repository is exactly [equation \(1.1\)](#). Its known pointwise eigen-identities are strengthened to a full spectral theorem on H_R , with algebraic multiplicities and a proof that nonperiodic analytic functions contribute no additional nonzero spectrum. The general theorem also identifies precisely when that last assertion would fail: a nonzero boundary value $w(0)$ creates an infinite geometric spectral ladder.

The source’s RMS normalization is reduced to an analytic observable that belongs to the chosen space. Its $-1/2$ projection survives integration and has a rigorously enclosed negative coefficient. The rate is therefore a true leading correction, not simply a possible rate in a norm estimate. The fixed-shell bound certifies alternating sides of the limit from $n = 10$ onward.

11.2 What has not been claimed

The finite spectrum does not describe the operator on every regularity space. The trace-class disk norm is different from the uniform norm on all continuous functions, and scalar uniform convergence does not imply a uniform operator-norm spectral gap on $C[0, 1]$. The statement for noninteger powers $|\sin \pi x|^q$ is not covered by the finite trigonometric-polynomial argument. The integer-only certificate does not constitute a kernel-checked Lean proof. Finally, the literature review establishes relevant classical context and a precise repository gap, not a definitive first-publication priority for every generalization.

11.3 Dependency and evidence ledger

Result	Basis	Status in this package
Fredholm determinant facts	Classical trace-class spectral theory [4, 5]	Explicitly stated standard input
General factorization	Affine trace lemma and roots-of-unity filtering	Full conventional proof
Quadratic complete spectrum	Exact 3×3 matrix and general theorem	Full conventional proof
Projection recursions and remainder	Fourier halving, jet gain, Cauchy bounds	Full conventional proof
RMS transfer identity	Exact sinc-product splitting and branch substitution	Full conventional proof
$B < 0$ and numerical enclosures	Proven analytic tails plus integer-only finite sums	Executed computer-assisted certificate
Finite matrices and mode identities	Rational symbolic calculations	Executed checks; not substitutes for general proofs
New Lean/Rocq declarations	None supplied or built	Not claimed

11.4 A practical formalization sequence

The shortest route into the repository begins with the finite algebra, not with a new spectral library. First formalize [equation \(5.2\)](#), the sequences η, α , and the exact mode theorem. The existing `RMSTransferEigenfunctions` identities already match the base case. Next formalize the endpoint-difference identity and its derivative-order consequence. A verified checker for the finite certificate can be kept small: fixed-point interval operations, alternating-series remainders, positive-series tails, and exact rational comparisons suffice.

The analytic bridge then has two possible routes. The spectral route formalizes trace-class weighted composition, the affine fixed-point trace, and the determinant theorem. The constructive route formalizes the Fourier remainder estimate and applies it directly to the shell identity. The latter can establish the RMS expansion without first building a complete Fredholm determinant API. Both routes should retain the explicit function-space hypotheses and the distinction between integer certificates and unverified floating-point computations.

12 Further research questions and topics

The following questions go beyond what is proved here. They are proposed research targets, not assertions of open status in the entire literature.

Question 12.1 (Sharp speed of the quasinilpotent remainder). Is the coefficient $\frac{1}{2} \log 2$ in the uniform upper bound [equation \(6.6\)](#) optimal for a natural bounded class of analytic observables? Can the stronger damping visible in repeated endpoint differences improve it? A useful result would give matching upper and lower bounds for an explicit nonperiodic observable, rather than only an improved constant in a general estimate.

Question 12.2 (Spectra at finite smoothness). Determine the complete spectrum or essential spectral radius of $\mathcal{L}_2$ on C^r , Hölder, Sobolev, and weighted endpoint spaces. Which analytic eigenvalues remain isolated, and how do unmatched endpoint jets interact with the noncompact part? The exact boundary intertwining relation provides a starting point, but the analytic determinant cannot simply be transferred.

Question 12.3 (Even-power matrix structure). For the integer matrices B_m , classify realness, simplicity, interlacing, and the ordering of nonzero eigenvalue moduli as m grows. Determine when the forced outer-row eigenvalue $(-1)^m$ resonates with the interior block and whether nontrivial Jordan chains occur. The exact polynomials in the package supply initial data, not an induction proof.

Question 12.4 (Large-moment asymptotics). Use the finite matrices to derive asymptotic expansions for the leading and subleading eigenvalues as $m \rightarrow \infty$. Relate these expansions to the maximizing periodic orbit of the sine product, and quantify the transition from finite even moments to the extremal envelope.

Question 12.5 (Noninteger powers and the arithmetic mean). For $q \notin 2\mathbb{N}$, establish certified subleading spectra and nonzero observable corrections for the transfer weight $|\sin \pi x|^q$. The case $q = 1$ is particularly relevant to the repository's arithmetic-mean decay. A suitable Banach space must accommodate the endpoint regularity and zeros; finite Fourier truncation alone is no longer exact.

Question 12.6 (Boundary perturbations and resonances). For a trigonometric polynomial family w_ε with $w_0(0) = 0$ and $w_\varepsilon(0) \neq 0$, the determinant theorem explicitly creates a geometric ladder. Describe the spectral projections and Jordan structure when one of these ladder values

meets a finite-band eigenvalue. Can one obtain uniform perturbation expansions through a resonance?

Question 12.7 (Infinite Fourier weights). Extend the boundary-factor formula to analytic weights with infinitely many Fourier coefficients by giving an explicit determinant truncation error. The finite-band theorem suggests an exact boundary factor and a controlled infinite periodic component. Such a result could turn heuristic spectral collocation into an a priori certified method.

Question 12.8 (Higher-dimensional boundary corrections). For expanding integral matrices acting on a cube or parallelepiped, determine the analogues of the finite Fourier band and the repeated-endpoint factor. One expects distinct contributions from faces and their intersections. A correct theorem must distinguish the interval or cube operator from the corresponding torus operator.

Question 12.9 (Arithmetic and regularity of the correction functional). The integer sequence α has $\alpha(2^j) = (-2)^j$, so its linear growth bound is sharp. Determine optimal regularity classes on which $g \mapsto \sum_{k \neq 0} \alpha(|k|) \hat{g}(k)$ extends continuously, and identify any useful distributional, refinement, or arithmetic interpretation of this functional.

Question 12.10 (Verified constants and complete repository integration). Produce Lean or Rocq proof objects for the finite certificate and the constructive Fourier remainder theorem. Increase the rigorously enclosed digits of A and B , and certify the earliest shell index at which the alternating sign persists. The theorem here proves the cutoff ten but does not claim it is minimal.

13 Conclusion

The missing implication in the repository was not an additional numerical eigenvalue calculation. It was a bridge from a finite Fourier computation to the full analytic interval operator, followed by a check that the resulting spectral correction actually occurs in the Rvachev observable. The boundary-factor theorem supplies the first bridge, and the exact Fourier projection recursion with integer enclosures supplies the second.

The resulting conclusion is stronger than the originally numerical rate: there are no other nonzero analytic spectral values in the quadratic case, the remainder after the two spectral terms has an explicit square-exponential upper bound, and the normalized RMS is provably on alternating sides of its limit from shell ten onward. The same mechanism gives exact finite-dimensional spectral problems for every even sine power and clarifies when nonperiodic boundary modes must instead be retained.

A Repository provenance and reproducibility

A.1 Inspected repository source

Repository: <https://github.com/VladimirReshetnikov/ProveIt>. The comparison uses commit `de4ac5ab1438bcbeed58481c4449b593074a1a1f`, observed on 28 September 2026 in Pacific time (29 September in UTC). The canonical article's Git blob is `aeba04f6c5f2d1eccff67645f773ba71803fa2a7`. Its path is

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/
rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay/
Rvachev_Up_Fourier_Decay.tex
```

The immutable source link is provided in the bibliography as [1]. The sibling README identifies the canonical synthesis and distinguishes it from a separately supplied rewrite. Source lines 1551–1555 state the unresolved spectral/RMS claim. The source’s explicit eigenfunctions and RMS power are treated as antecedents, not as new contributions of this article. No repository file was modified during preparation of this package.

A.2 Files and commands

The package contains this source and its compiled PDF, the primary integer-only certificate, an independent interval-arithmetic cross-check, an exact algebra checker, their executed outputs, a provenance file, and build instructions. The minimal numerical verification needs only Python:

```
python certify_integer.py --output integer_certificate.json
```

The optional cross-check and finite symbolic checks use the versions listed in `requirements.txt`:

```
python certify_constants.py --output constants_certificate.json
python verify_algebra.py --output algebra_certificate.json
```

The article has an internal bibliography and no external figure dependencies. It can be rebuilt by running

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

All assertions in the checkers are part of the intended execution: do not run Python with its `assertion-disabling` optimization flag.

A.3 Finite checks performed

The algebra checker verifies the characteristic polynomials of $4^m A_m$ for $1 \leq m \leq 5$, the semisimplicity of the quadratic nonzero spectrum, and the double quartic value $1/16$. It checks 80 finite root-of-unity trace identities across $2 \leq b \leq 5$, $0 \leq M \leq 4$, and $1 \leq n \leq 4$. It also verifies the exact finite-mode formula for $1 \leq k \leq 512$. These checks detect indexing and normalization mistakes; the proofs in the main text, not the finite test ranges, establish the general statements.

References

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